

# Tree Coin Collecting I

Time limit: 1.00 s      Memory limit: 512 MB

You are given a tree with  $n$  nodes. Some nodes contain a coin.

Your task is to answer  $q$  queries of the form: what is the shortest length of a path from node  $a$  to node  $b$  that visits a node with a coin?

## Input

The first line contains two integers  $n$  and  $q$ : the number of nodes and queries. The nodes are numbered  $1, 2, \dots, n$ .

The second line contains  $n$  integers  $c_1, c_2, \dots, c_n$ . If  $c_i = 1$ , node  $i$  has a coin. If  $c_i = 0$ , node  $i$  doesn't have a coin. You can assume at least one node has a coin.

Then there are  $n - 1$  lines describing the edges. Each line contains two integers  $a$  and  $b$ : there is an edge between nodes  $a$  and  $b$ .

Finally, there are  $q$  lines describing the queries. Each line contains two integers  $a$  and  $b$ : the start and end nodes.

## Output

Print  $q$  integers: the answers to the queries.

## Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

## Example

| Input                                                                    | Output           |
|--------------------------------------------------------------------------|------------------|
| 5 4<br>1 0 0 1 0<br>2 4<br>2 3<br>1 3<br>3 5<br>1 5<br>3 2<br>4 4<br>5 5 | 2<br>3<br>0<br>4 |